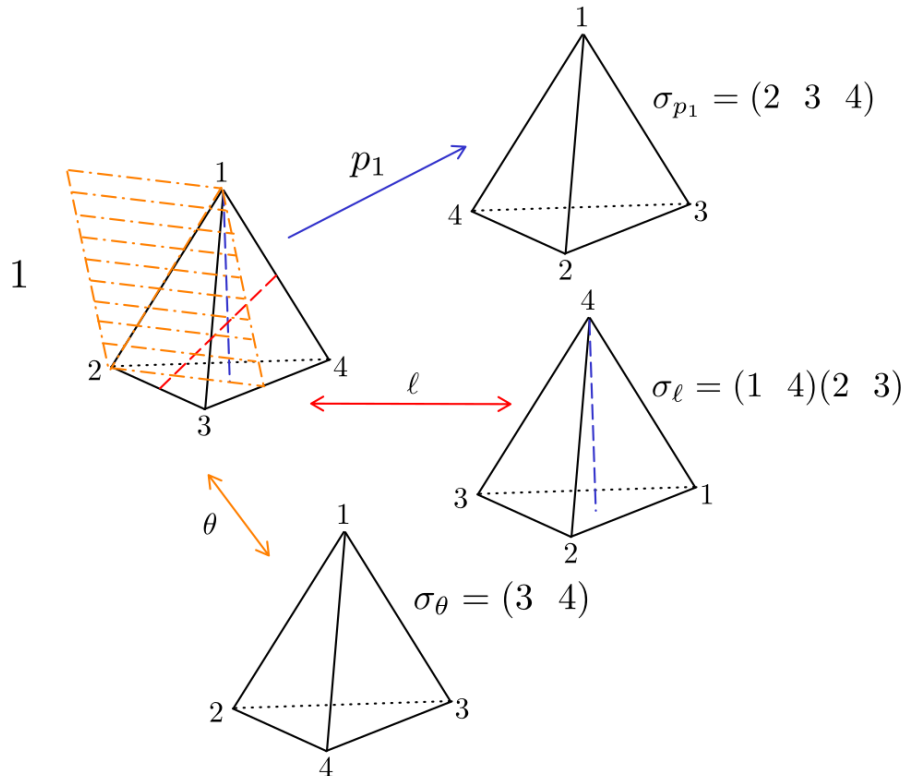




1. *True or false?* “I have solved first two tutorials”. Here, I denotes the person reading this and not the instructor.
2. The following picture depicts the symmetries p_1, ℓ and θ in terms of permutations. The way to write permutation σ_{p_1} is as follows: Since under the transformation p_1 , the vertex 1 does not change its location, we do not write it in σ_{p_1} . The vertex 2, under the transformation p_1 occupies the original position of the vertex 3, so we write (2 3. Along the same lines, the vertex 3 occupies the position of the vertex 4, and we continue writing (2 3 4. The vertex 4 occupies the position of vertex 2 – so a cycle is complete, and we close the bracket to write (2 3 4). There are no more vertices and thus the permutation corresponding to p_1 is $\sigma_{p_1} = (2\ 3\ 4)$.



- (a) Show that $\sigma_{\theta p_1} = \sigma_{\theta} \sigma_{p_1}$.
- (b) Pick two arbitrary $\alpha, \beta \in \text{sym}(\mathbb{T})$. Convince yourself that $\sigma_{\beta\alpha} = \sigma_{\beta} \sigma_{\alpha}$.
- (c) Following the notation of class notes, write

$$\mathcal{A} := \{1, p_1, p_1^2, p_2, p_2^2, p_3, p_3^2, p_4, p_4^2, \ell, \ell', \ell''\},$$

and

$$\theta\mathcal{A} := \{\theta, \theta p_1, \theta p_1^2, \theta p_2, \theta p_2^2, \theta p_3, \theta p_3^2, \theta p_4, \theta p_4^2, \theta\ell, \theta\ell', \theta\ell''\}.$$

Recall that $\text{sym}(\mathbb{T}) = \mathcal{A} \cup \theta\mathcal{A}$. Use the permutation notation and composition of permutations to identify six symmetries in $\theta\mathcal{A}$ which are not reflections.

3. Which of the following form a subgroup of $\text{sym}(\mathbb{T})$?

(a) $\{1, p_1, p_1^2, p_2, p_2^2, p_3, p_3^2, p_4, p_4^2\}.$

(b) $\{1, \ell, \ell', \ell''\}.$

(c) $\{1, p_1, p_1^2, p_2, p_2^2, p_3, p_3^2, p_4, p_4^2, \ell, \ell', \ell''\}.$

4. Identify as many subgroups of $\text{sym}(\mathbb{T})$ as you can.

5. Let S denote the set of mirror symmetries of $\mathbb{T}$. Show that S is a generating set of $\text{sym}(\mathbb{T})$.
